Haoxue Ran

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EDUCATION

Xi'an University of Architecture and Technology **Sep 2021 – Expected Jun 2027**
Ph.D., Environmental Engineering (*Direct-Track, "Cuiying" Excellence Program*)
Advisor: Prof. Rui Miao
Additional Guidance: Prof. Jin Zhao, Xi'an Jiaotong University

Guilin University of Technology **Sep 2017 – Jun 2021**
B.E., Environmental Engineering (*Elite Engineering Program*)
Academic Rank: 1/30 within the program

RESEARCH EXPERIENCE

Graduate Student Researcher **Xi'an University of Architecture and Technology**
Advisor: Prof. Rui Miao **2022 – Present**
Department of Environmental and Municipal Engineering

- Developed a trace-dose Fenton pretreatment technology at neutral pH to mitigate membrane organic fouling, and elucidated its underlying mechanisms.
- Established a systematic analytical framework based on data-driven models to predict membrane fouling behavior and identify its key governing factors.
- Quantitatively decoupled the contributions of individual causal pathways to membrane fouling mitigation by integrating causal inference with micro-interfacial analysis techniques.
- Investigated the different mechanisms of divalent metal ions (Ca^{2+}, Mg^{2+}) on the efficacy of Fenton pretreatment in membrane organic fouling mitigation.
- Quantitatively decoupled the key factors and causal relationships of Ca^{2+}/Mg^{2+} ions governing Fenton pretreatment for membrane fouling mitigation by integrating artificial intelligence with molecular dynamics simulations.

PUBLICATIONS & PATENTS

Journal Papers

1. **Ran, H.**, Miao, R.*, Zhao, J.*, Zhang, H., Wang, S., Zhang, J., Wang, P., Wang, L. and Yin, J. **2026**. Quantitative decoupling of the key factors and mechanisms of Fenton-pretreated membrane fouling mitigation via machine learning-assisted causal inference. *Water Research*, 303, 126247.
2. **Ran, H.**, Wang, Y., Miao, R.*, Zhao, J.*, Wang, S., Zhang, J., Wang, P., Wang, L. and Yin, J. **2026**. Effects of feature selection on predictive performance and interpretability in data-driven modeling:

A case study of membrane flux prediction after Fenton pretreatment. *Separation and Purification Technology*, 393, 137304.

3. Miao, R.*, **Ran, H.**, Yang, Y., Li, Y., Ma, Z., Lv, Y., Meng, X., He, M. and Wang, L. **2024**. *In situ* acid production by organic matter induced with trace homogeneous Fenton reagent for membrane fouling control. *Water Research*, 258, 121752.
4. Miao, R.*, Wang, Y., **Ran, H.**, Li, Y., Yang, C. and Wang, L. **2023**. Relative importance of drag force and adhesion forces in the fouling of ultrafiltration membranes by secondary effluent organic matter after pre-ozonation. *Journal of Membrane Science*, 678, 121677.

Manuscripts in Preparation

- Differential mechanisms of Ca^{2+} and Mg^{2+} ions governing Fenton pretreatment for membrane organic fouling mitigation.
- Decoupling the key factors and causal relationships of Ca^{2+}/Mg^{2+} governing Fenton pretreatment for membrane fouling mitigation via artificial intelligence and molecular dynamics simulations.

Patents & Software Copyrights

- Miao, R., **Ran, H.**, Wang, L., Lv, Y., Yang, C. and Wang, Y. (**2023**). *Method and application for membrane fouling control via Fe^{2+}-H_2O_2 induced in situ acid production from organic pollutants*. Chinese Invention Patent, Application No: CN202310381227.4.
- Miao, R., **Ran, H.**, Zhao, C., Wang, S., Lei, Y. and Zhang, J. (**2025**). *Method for membrane fouling mechanism identification and regulation based on causal inference*. Chinese Invention Patent, Application No: CN202511264800.9.
- Miao, R., **Ran, H.**, Yang, Y., Wang, S., Lei, Y. and Zhang, J. (**2025**). *Machine learning-based method and system for the optimal control of membrane fouling pretreatment*. Chinese Invention Patent, Application No: CN202510539314.7.
- Miao, R., **Ran, H.**, Wang, Y., Wang, S., Lei, Y. and Zhang, J. (**2025**). *Method and system for membrane fouling prediction and key factor identification/regulation in Fenton pretreatment*. Chinese Invention Patent, Application No: CN202511401234.1.
- Bai, S., **Ran, H.**, Wu, H. and Zhou, P. (**2020**). *Intelligent maintenance device for green building vegetation*. Chinese Utility Model Patent, Patent No: ZL202020307040.1.
- **Ran, H.**, Miao, R., Wang, S., Lei, Y. and Zhang, J. (**2025**). *Dosage optimization system for Fenton pretreatment-based membrane fouling mitigation under various operating conditions*. Chinese Computer Software Copyright, Registration No: 2025SR2214520.

RESEARCH PROJECTS & GRANT PROPOSALS

National Natural Science Foundation of China (NSFC, General Program)

Mechanisms and mitigation strategies of inorganic salt crystallization fouling on ultrafiltration membranes during lithium extraction from salt lake brine.

Role: Co-drafted proposal. **2027 – 2030**

Natural Science Basic Research Program of Shaanxi Province (Youth Project, Class A)
Research on theory and application of membrane fouling control based on multidimensional influencing factors.
Role: Co-drafted proposal & Graduate Research Assistant. **2026 – 2029**

National Key Research and Development Program of China (Sub-project)
Interfacial migration mechanisms of salt ions in brine and fouling prevention/control technologies for membranes/adsorbents.
Role: Graduate Research Assistant. **2022.10 – 2026.09**

National Natural Science Foundation of China (NSFC, General Program)
Mechanism and regulation of hydration repulsion for membrane fouling mitigation during ultrafiltration.
Role: Graduate Research Assistant. **2021.01 – 2024.12**

Youth Innovation Team of Shaanxi Universities
Innovative Team of Membrane Separation Technology for Valuable Resource Extraction from Salt Lake Brine.
Role: Graduate Research Assistant. **2023.01 – 2026.12**

International Science and Technology Cooperation Project of Shaanxi Province
High-efficiency reuse process of municipal secondary effluent centered on ultrafiltration technology.
Role: Graduate Research Assistant. **2022.01 – 2023.12**

National Natural Science Foundation of China (NSFC, General Program)
Self-cleaning mechanism and regulation of microorganisms on MBR membrane surface induced by self-forming dynamic membranes.
Role: Research Contributor. **2024.01 – 2027.12**

Enterprise-funded Collaborative R&D Projects
Resource utilization of kitchen and septic tank domestic wastes in Shendong Coal.
Role: Research Contributor. **2022.07 – 2026.12**

Recycled utilization of mine water for promoting low-carbon emission reduction in boilers.
Role: Research Contributor. **2025.05 – 2025.11**

HONORS & AWARDS

Research & Innovation Awards

- Grand Prize, China Youth Science and Technology Innovation Challenge 2026
- Third Prize, 19th "Challenge Cup" National Undergraduate Academic Science and Technology Competition 2025
- Best Presentation Award, 20th National Graduate Environmental Forum (Tongji University) 2025
- Outstanding Undergraduate Thesis Award, Guilin University of Technology 2021

- Third Prize, 16th "Challenge Cup" National Undergraduate Academic Science and Technology Competition 2019
- Grand Prize, "Challenge Cup" National Undergraduate Academic Science and Technology Competition (*Guangxi Regional Selection*) 2019

Scholarships & Honors

- National Scholarship for Graduate Students, Ministry of Education of China 2026
- Second-class Scholarship of Xi'an University of Architecture and Technology 2026
- "Yu Dan" Corporate Excellence Scholarship, Xi'an University of Architecture and Technology 2025
- Second-class Scholarship of Xi'an University of Architecture and Technology 2025
- Second-class Scholarship of Xi'an University of Architecture and Technology 2024
- First-class Scholarship of Xi'an University of Architecture and Technology 2023
- Second-class Scholarship of Xi'an University of Architecture and Technology 2022
- Guangxi Provincial Outstanding Graduate Honor 2021
- Guangxi Provincial Government Scholarship 2020
- Special-class Scholarship of Guilin University of Technology 2020
- Outstanding Student Award, Guilin University of Technology 2020
- First-class Scholarship of Guilin University of Technology 2019
- Outstanding Student Award, Guilin University of Technology 2019
- First-class Scholarship of Guilin University of Technology 2018
- Elite Engineer Fellowship, Guilin University of Technology 2018
- Outstanding Student Award, Guilin University of Technology 2018

RESEARCH PRESENTATIONS

1. "Decoupling Fenton-Mitigated Membrane Fouling Mechanisms via Causal Inference and Machine Learning," *8th IWA Regional Membrane Technology Conference (IWA-RMTC 2026)*, Shanghai, China, Aug 2026. (Oral Presentation).
2. "Deciphering Complex Membrane Fouling Mechanisms via Explainable Machine Learning: A Case Study of Fenton-Pretreated Membranes," *Academic Exchange on Green Low-Carbon and Sustainable Development (IEEPA Annual Meeting)*, Hangzhou, China, Dec 2025. (Oral Presentation).
3. "Mechanisms of Trace Fe^{2+}/H_2O_2 Pretreatment on Organic Fouling Behaviors of Ultrafiltration Membranes," *2025 China Membrane Industry Intelligent Manufacturing and Application Technology Conference*, Xi'an, China, Nov 2025. (Oral Presentation).
4. "Mechanisms of Trace Fe^{2+}/H_2O_2 Pretreatment on Organic Fouling Behaviors of Ultrafiltration Membranes," *20th National Graduate Environmental Forum (Tongji University)*, Shanghai, China,

May 2025. (Oral Presentation, **Best Presentation Award**).

5. "Mechanisms of Trace Fe^{2+}/H_2O_2 Pretreatment on Organic Fouling Behaviors of Ultrafiltration Membranes," *19th National Environmental Ph.D. Academic Conference (Tsinghua University)*, Beijing, China, May 2025. (Oral Presentation & **Session Chair**).

TEACHING AND MENTORING EXPERIENCE

Graduate Teaching Assistant **Xi'an University of Architecture and Technology**

Course: *Membrane Separation Theory and Technology* (Instructor: Prof. Rui Miao) **Spring 2022**

- Assisted in course logistics, graded homework assignments, and provided feedback to the instructor (about 120 students).

Research Seminar Coordinator **Miao Research Group**

Shaanxi Membrane Separation Research Institute **2021 – Present**

- Scheduled academic topics and managed logistics for monthly large-scale research group seminars (about 20–30 participants).

Mentoring and Research Advising

- Jiayi Zhang (Undergraduate Researcher) 2022 – Present
- Yifan Yang (M.S. Student) 2022 – 2025
- Siyu Wang (M.S. Student) 2023 – 2026
- Yuqing Lei (M.S. Student) 2024 – Present
- Fangru Zhu (M.S. Student) 2025 – Present

PROFESSIONAL SERVICE AND LEADERSHIP

Assisted Review: *Environmental Science & Technology*, *Water Research*, *Journal of Membrane Science*, *Chemical Engineering Journal*, *Desalination*, *Separation and Purification Technology*, *Journal of Water Process Engineering*.

Student Laboratory Manager: Miao Research Group, Shaanxi Membrane Separation Research Institute.

Safety & Operations Coordinator: Department of Environmental and Municipal Engineering, Xi'an University of Architecture and Technology.

RESEARCH SKILLS

Analytical Characterization: LC-TOF-MS, QCM-D, CLSM, LC-OCD, SEM, ICP-MS, AFM, FTIR, zeta-potential and hydrodynamic diameter measurements, and laser diffraction particle-size analysis.

Modeling & Computation: CatBoost, TabPFN, random forests, deep learning, explainable AI (XAI), causal inference and directed acyclic graphs (DAGs), and molecular dynamics (MD) simulations.